Lecture 7
The Network Layer
Data Plane

SDU

Subjects of today:

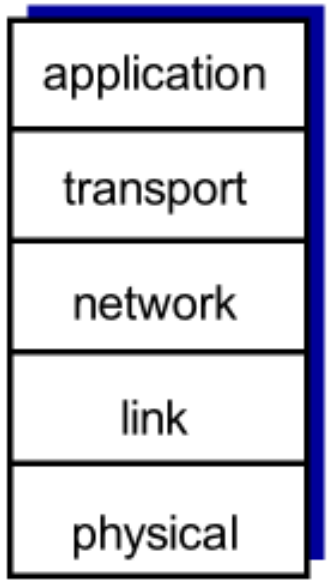
- The Role of the Network Layer
- The Router
- Internet Protocol Addressing
- Beyond IPv4, NAT and IPv6

SDU

7.1 The Role of the Network Layer

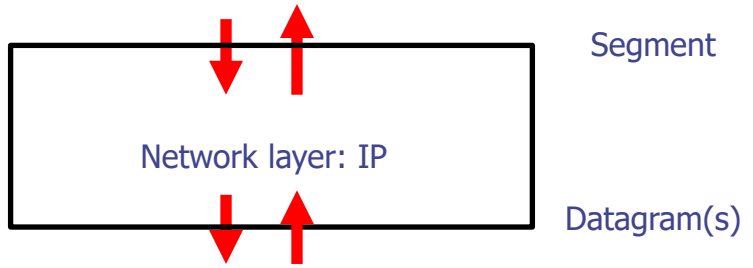
The Internet Protocol Stack

- *application:* supporting network applications
  - FTP, SMTP, HTTP
- *transport:* process-process data transfer
  - TCP, UDP
- *network:* routing of datagrams from source to destination
  - IP, routing protocols
- *link:* data transfer between neighboring network elements
  - Ethernet, 802.111 (WiFi), PPP
- *physical:* bits "on the wire"



Introduction 1-60

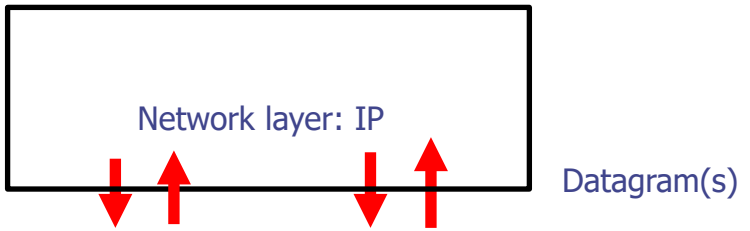
The Network Layer at the host…



- Retrieves and delivers segments from/to the Transport Layer.
- Comply with network layer protocols:
  - Data plane: IP (IPv4, IPv6).
  - Control plane: E.g. ICMP.
- Sends and receives datagrams to/from other layer 3 devices through the Link Layer.

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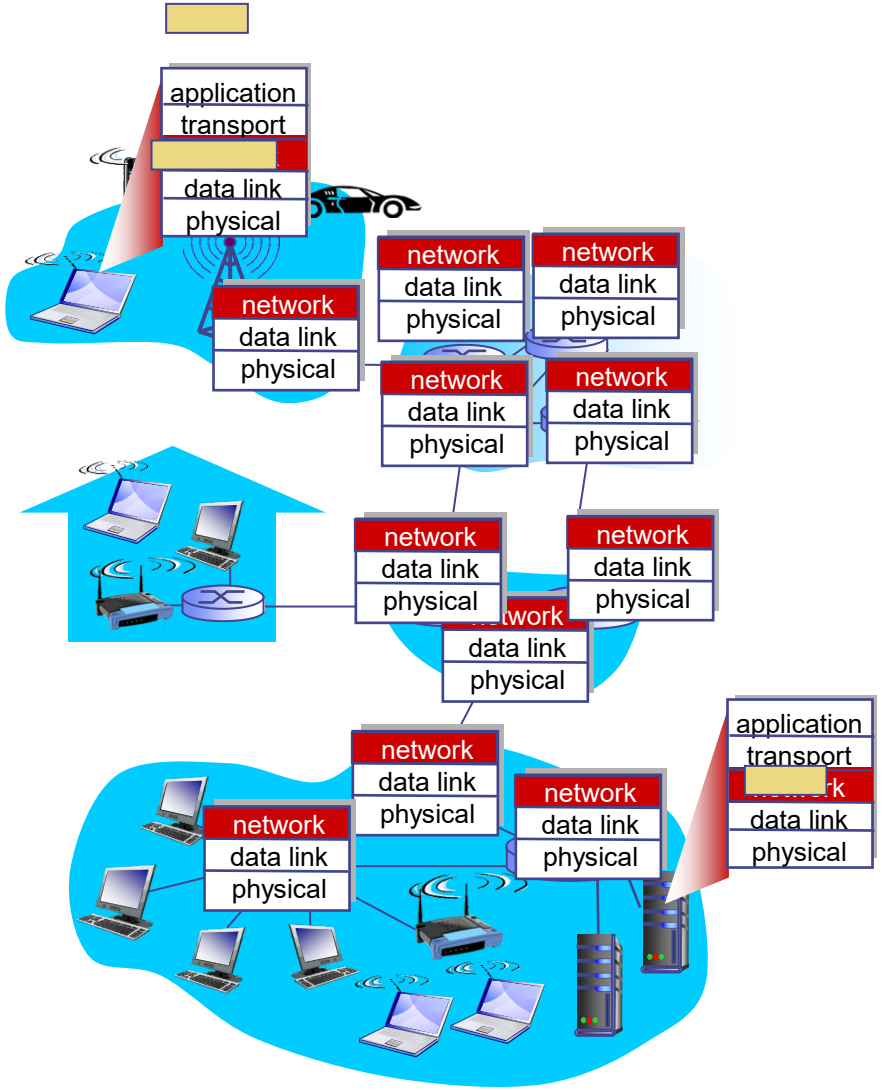
The Network Layer at the router…



- Comply with network layer protocols:
  - Data plane: IP (IPv4, IPv6).
  - Control plane: E.g. ICMP.
- Sends and receives datagrams to/from other layer 3 devices through the Link Layer.
- Forward datagrams from one interface (physical port) to another.
- Route datagrams to the "right" interface.

Network Layer

- transport segment from sending to receiving host
- on sending side encapsulates segments into datagrams
- on receiving side, delivers segments to transport layer
- network layer protocols in *every* host & router
- router examines header fields in all IP datagrams passing through it



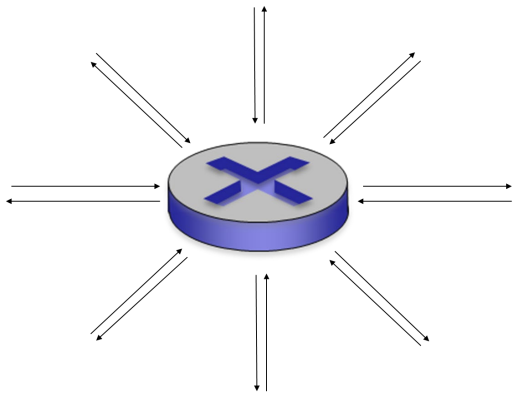
Two Key Network-Layer Functions

network-layer functions:

- forwarding: move packets from router's input to appropriate router output
- routing: determine route taken by packets from source to destination
  - routing algorithms

analogy: taking a trip

- forwarding: process of getting through single interchange
- routing: process of planning trip from source to destination

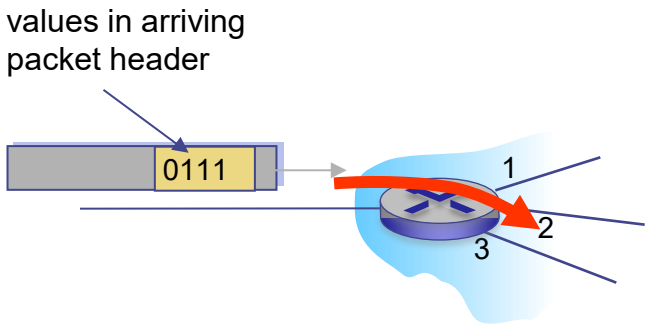


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Network layer: Data plane vs. Control plane

Data plane

- local, per-router function
- determines how datagram arriving on router input port is forwarded to router output port
- forwarding function

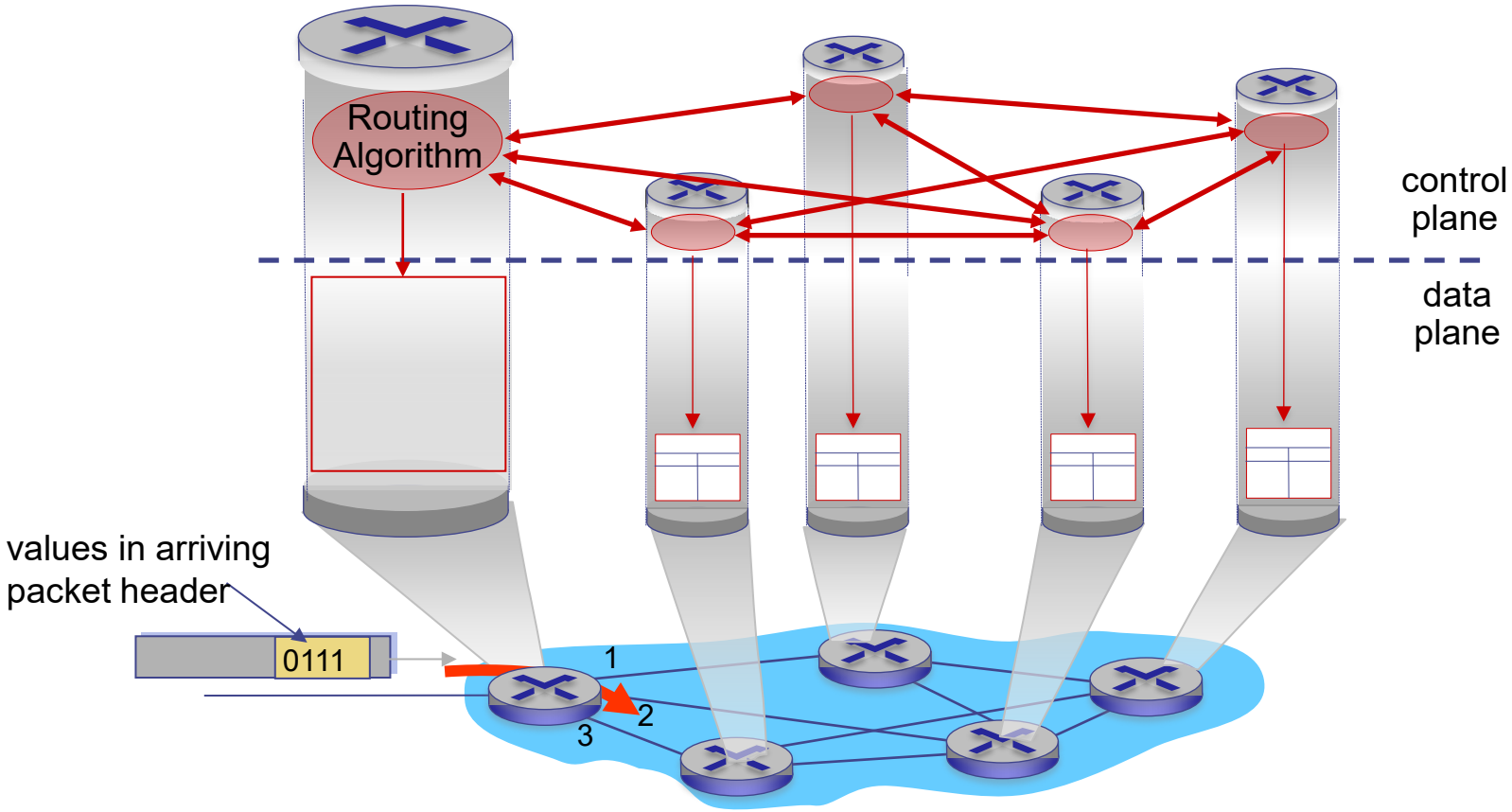


Control plane

- network-wide logic
- determines how datagram is routed among routers along end-2-end path from source host to destination host
- two control-plane approaches:
  - *traditional routing algorithms:* implemented in routers
  - *software-defined networking (SDN)*: implemented in (remote) servers

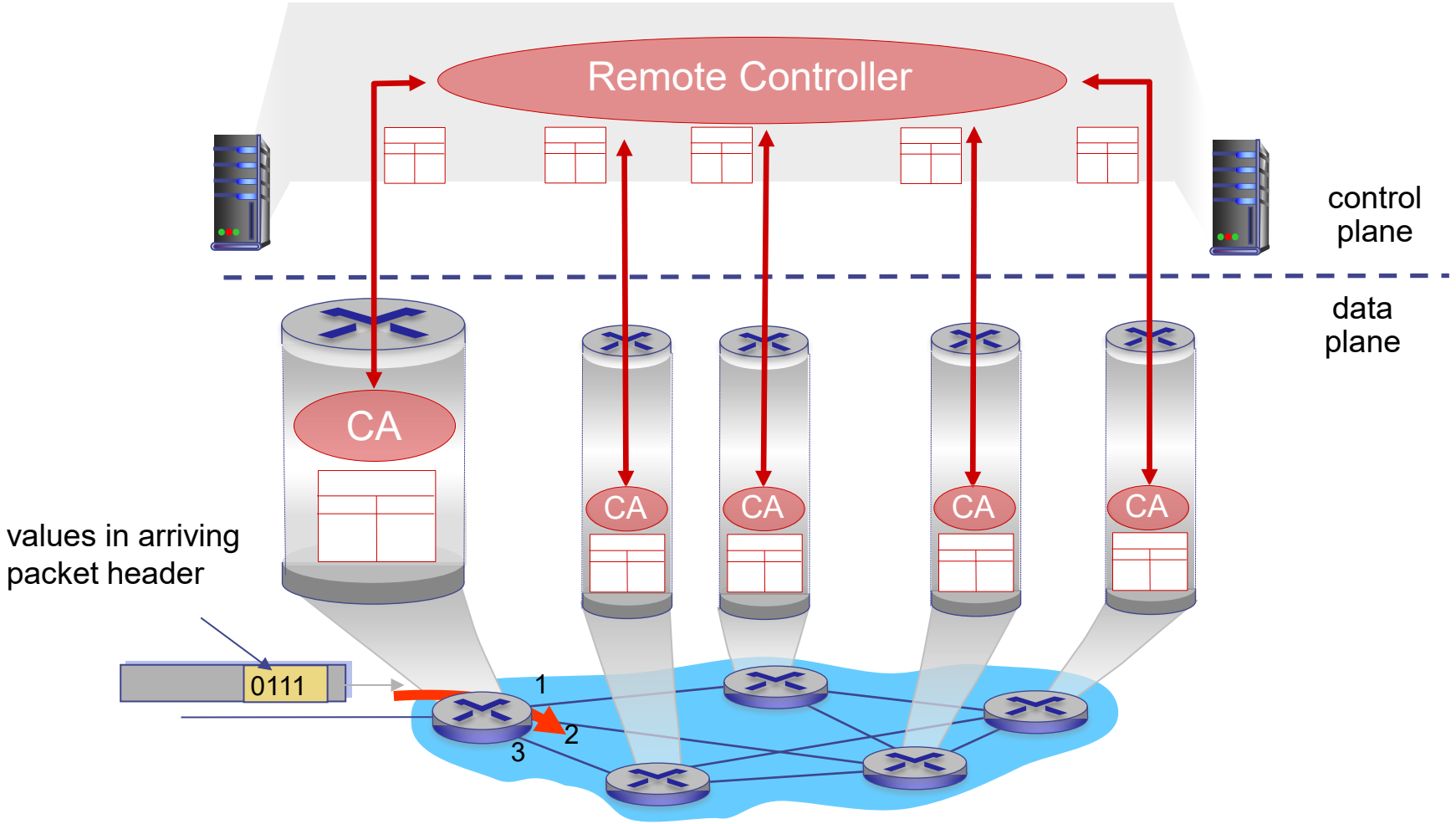
Per-router Control Plane

Individual routing algorithm components *in each and every router* interact in the control plane



Logically Centralized Control Plane

A distinct (typically remote) controller interacts with local Control Agents (CAs)



Network Service Model

Possibilities

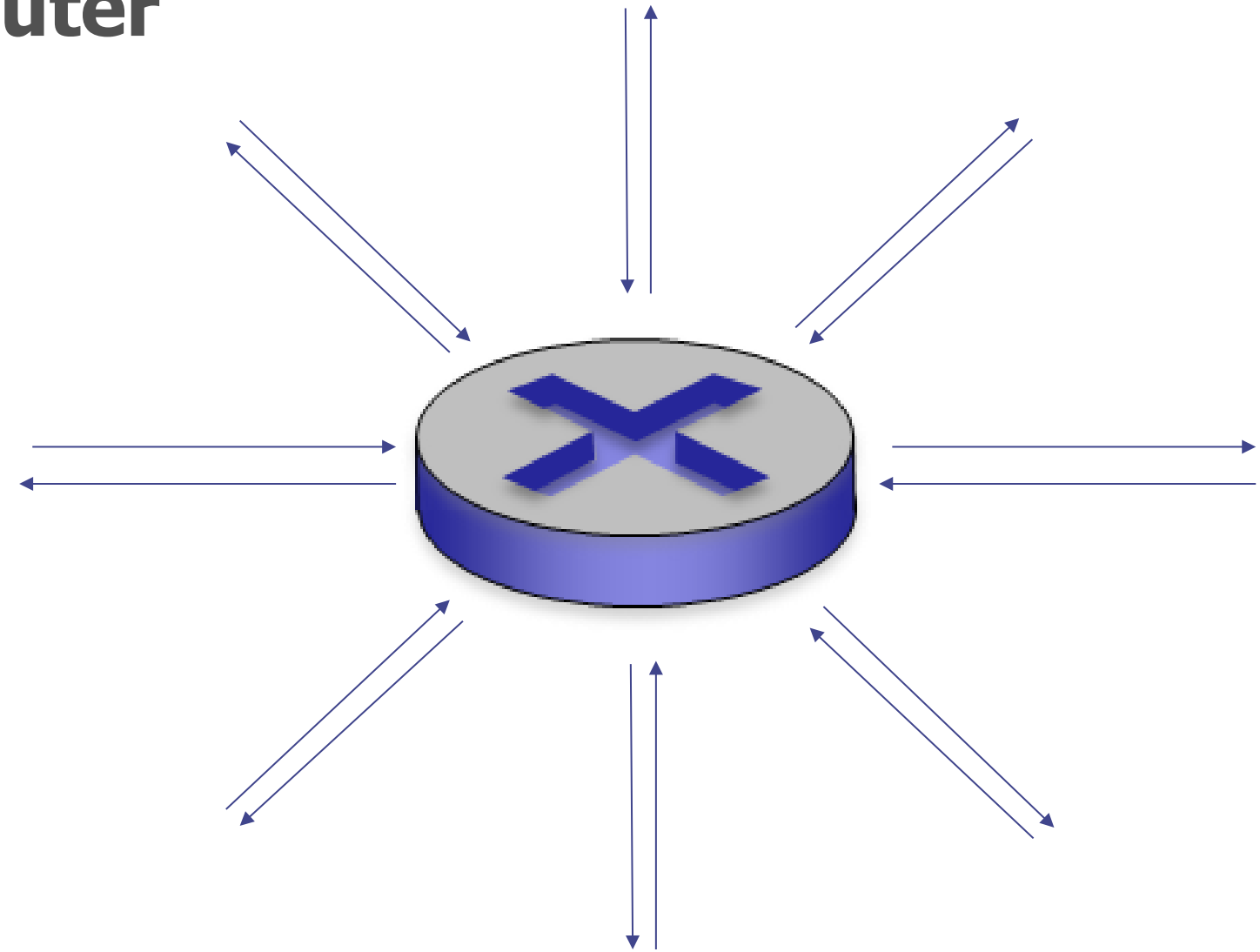
- Guarantee delivery (loss)
- Guarantee delivery with bounded delay (time)
- In-order packet delivery
- Ensure minimal bandwidth
- Ensure data integrity

Actual

- Best-effort service…

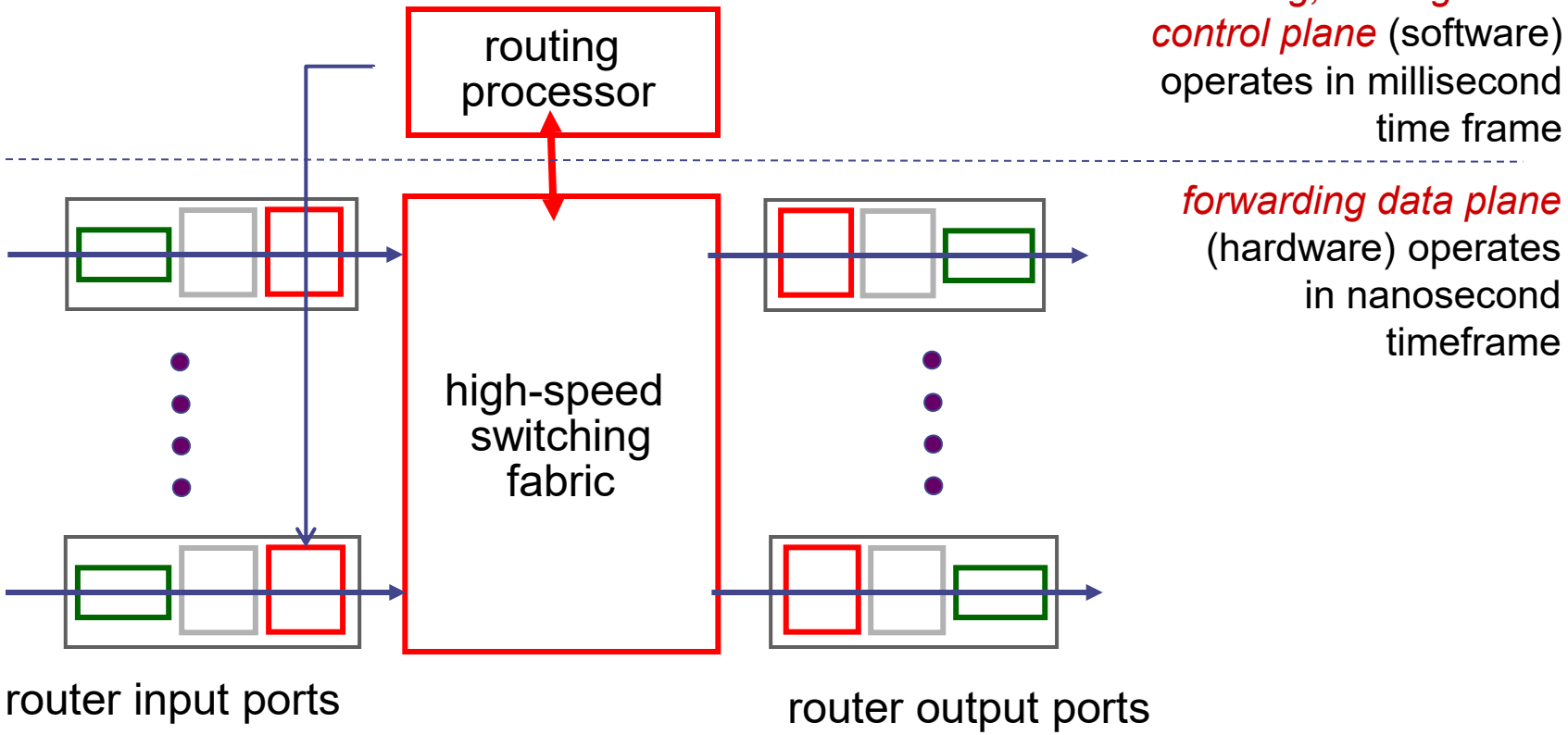
7.2 The Router

The Router

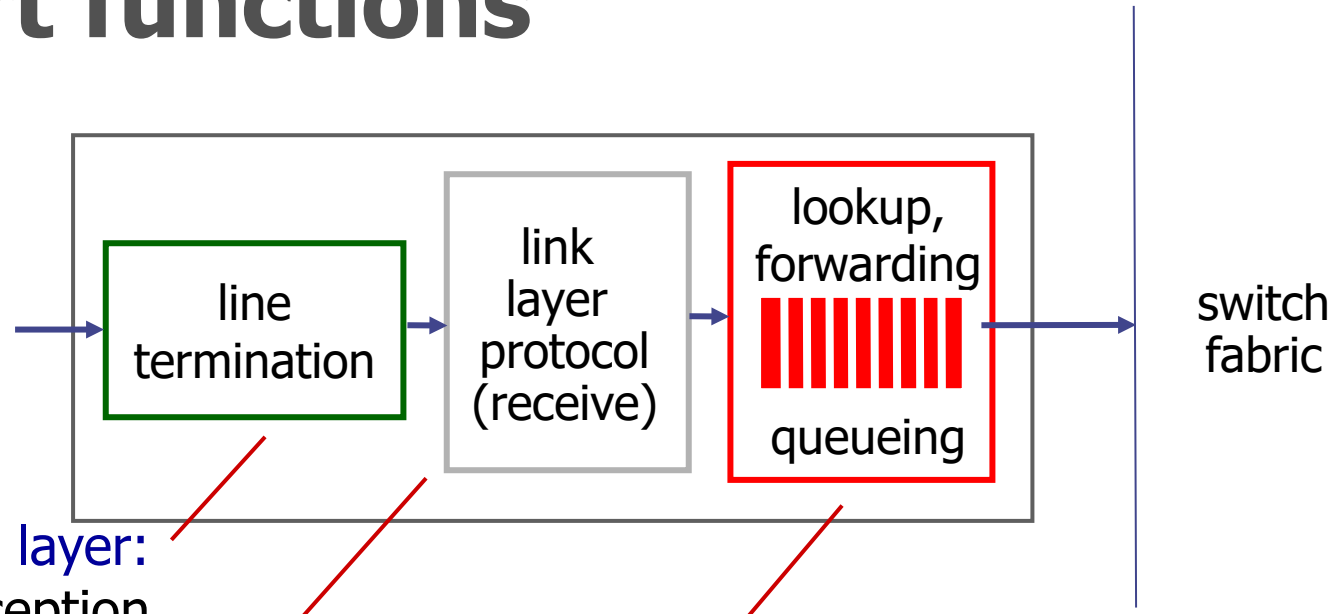


Router Architecture Overview

- high-level view of generic router architecture:



Input port functions



physical layer:
bit-level reception

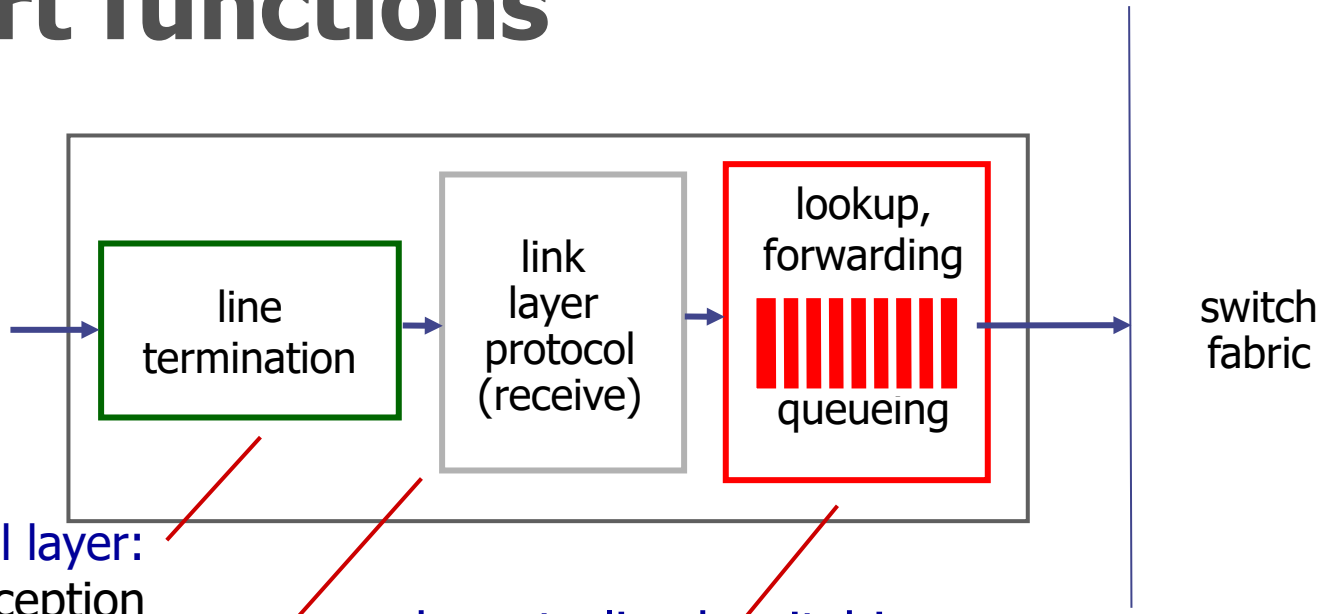
data link layer:
e.g., Ethernet
see chapter 6

decentralized switching:

- using **header field** values, **lookup** output port using forwarding table in input port memory *("match plus action")*
- **goal**: complete input port processing at 'line speed'
- queuing: if datagrams arrive faster than the forwarding rate into switch fabric

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Input port functions



physical layer:
bit-level reception

data link layer:
e.g., Ethernet
see chapter 6

decentralized switching:

- using header field values, lookup output port using forwarding table in input port memory (*"match plus action"*)
- *destination-based forwarding:* forward based only on destination IP address (traditional)
- *generalized forwarding:* forward based on any set of header field values

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Destination-based forwarding

IP address: 32-bit identifier

forwarding table

Destination Address Range	Link Interface
200.23.16.0 → 11001000 00010111 00010000 00000000 through **200.23.23.255** → 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011001 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

Q: but what happens if ranges don't divide up so nicely?

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Longest prefix matching

longest prefix matching

when looking for forwarding table entry for given destination address, use *longest* address prefix that matches destination address.

Destination Address Range	Link interface
11001000 00010111 00010*** *********	0
11001000 00010111 00011000 *********	1
11001000 00010111 00011*** *********	2
otherwise	3

examples:

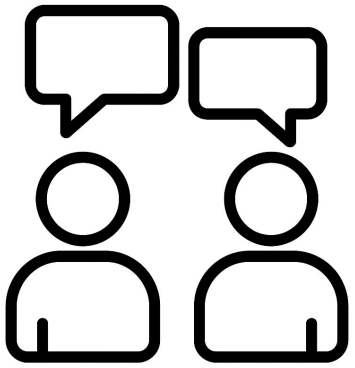
DA1: 11001000 00010111 00010110 10100001 which interface?

DA2: 11001000 00010111 00011000 10101010 which interface?

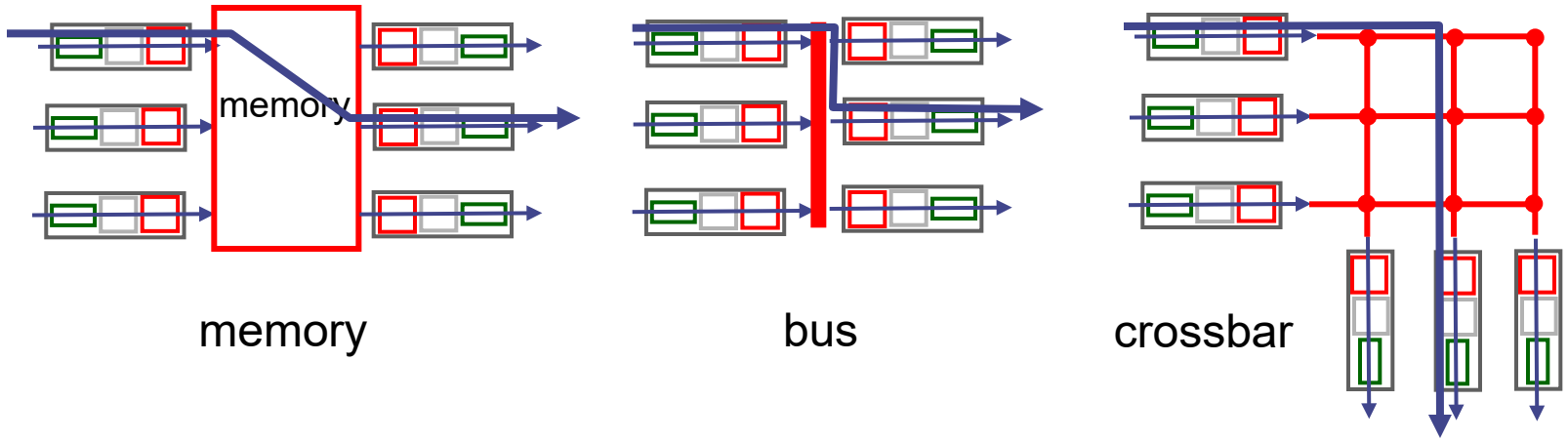
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Switching fabrics

- transfer packet from input buffer to appropriate output buffer
- switching rate: rate at which packets can be transfer from inputs to outputs
- three types of switching fabrics:

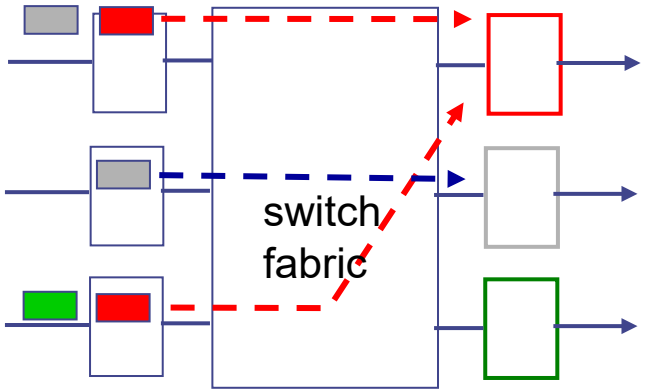


What are the drawbacks of each method?

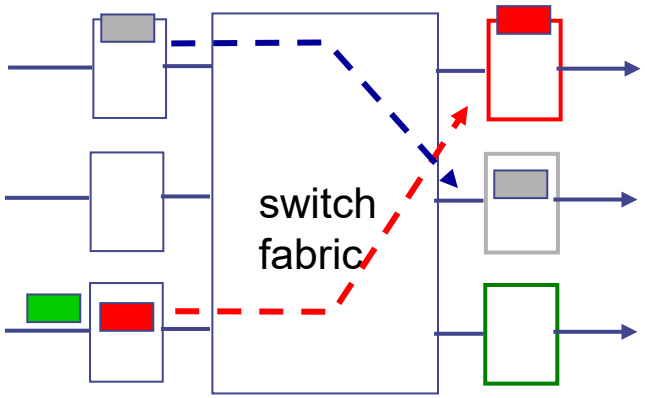


Input port queuing

- fabric slower than input ports → queueing may occur at input queues
  - *queueing delay and loss due to input buffer overflow!*
- Head-of-the-Line (HOL) blocking: queued datagram at front of queue prevents others in queue from moving forward



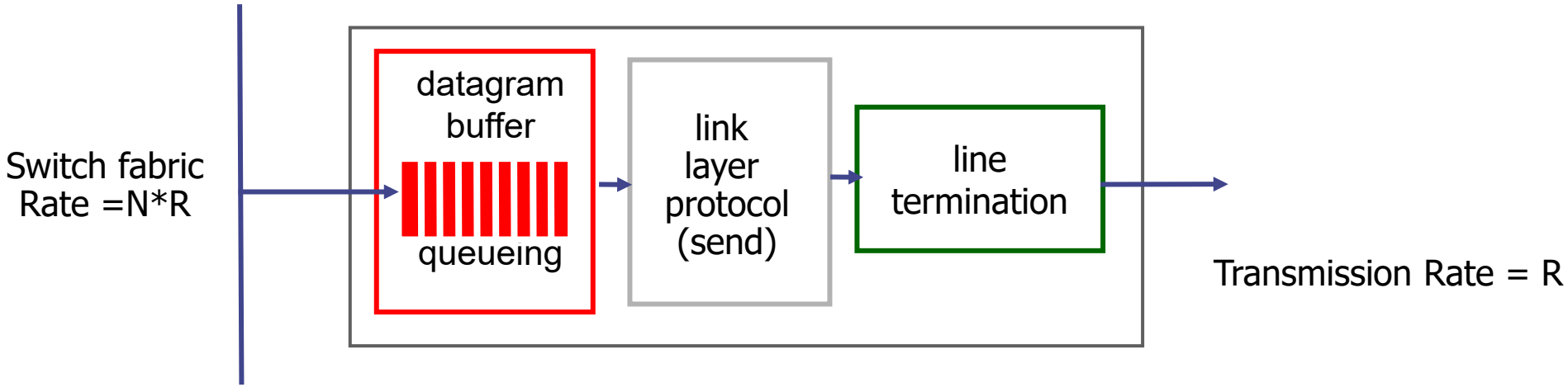
output port contention:
only one red datagram can be transferred.
lower red packet is blocked



one packet time later:
green packet
experiences HOL
blocking

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Output ports



- *buffering* required when datagrams arrive from fabric faster than the transmission rate

Datagram (packets) can be lost due to congestion/lack of buffers

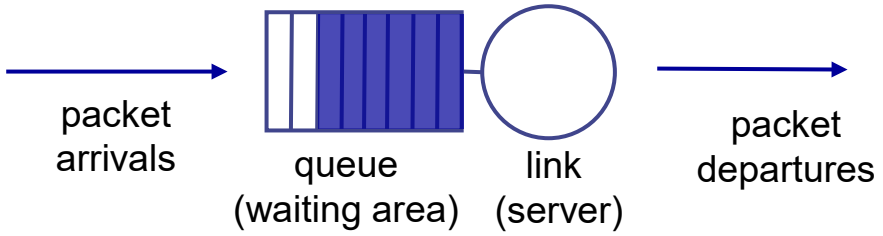
- *scheduling discipline* chooses among queued datagrams for transmission

Priority scheduling – who gets best performance?

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Scheduling Mechanisms

- Scheduling: choose next packet to send on link
- FIFO (first in first out) scheduling: send in order of arrival to queue
  - Discard policy: if packet arrives to full queue: who to discard?
    - Tail drop: drop arriving packet
    - Priority: drop/remove on priority basis
    - Random: drop/remove randomly

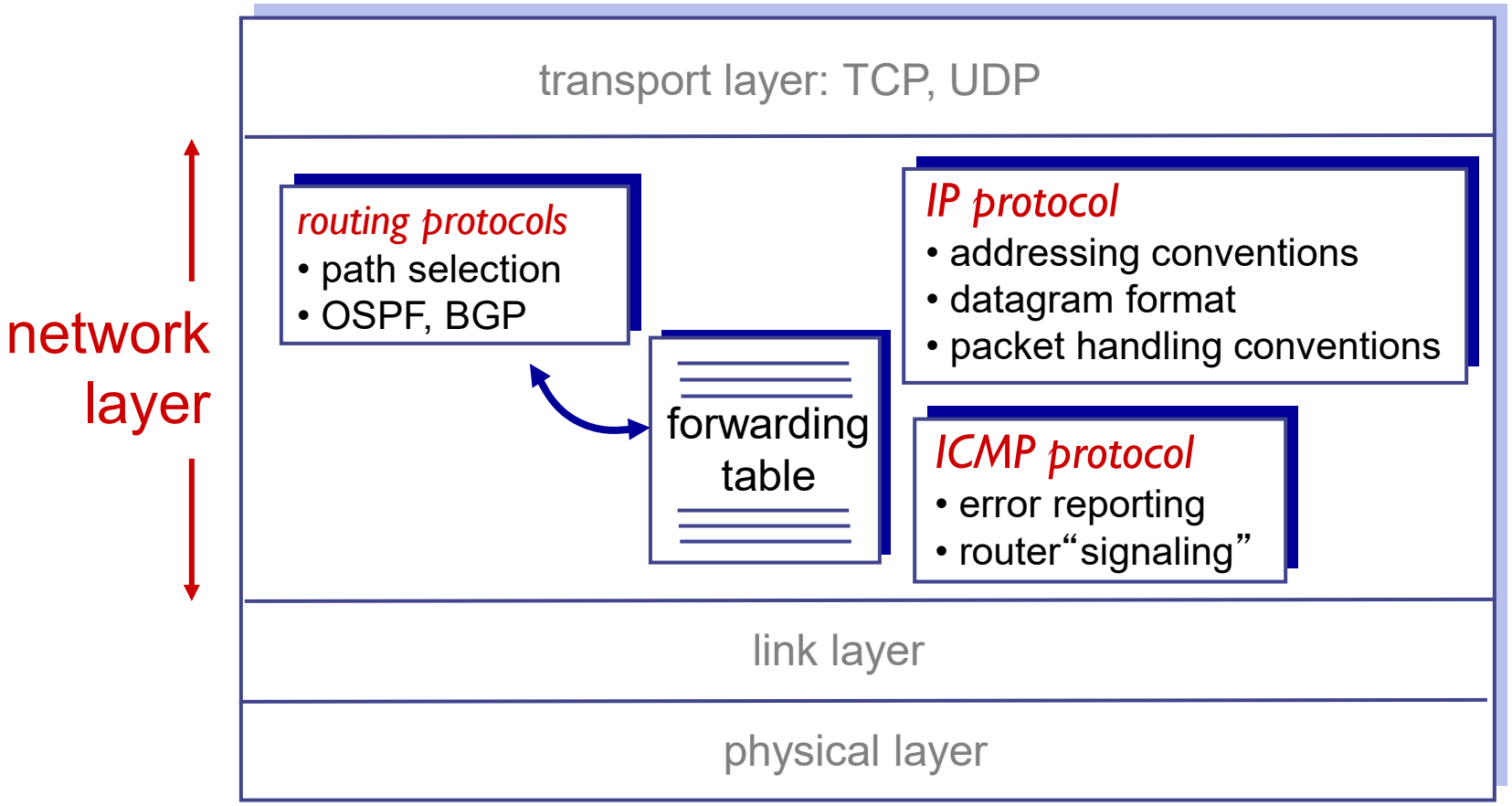


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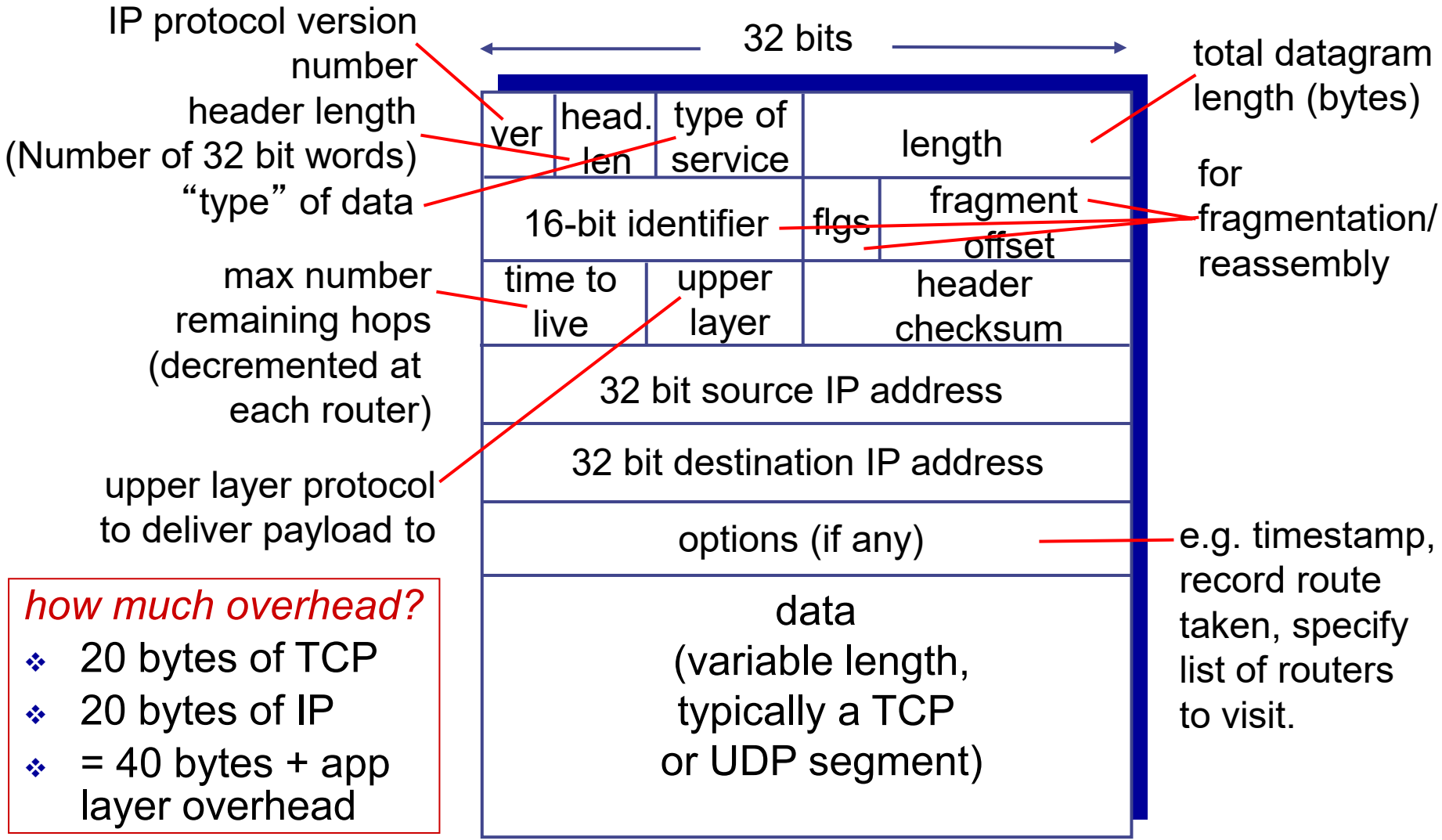
7.3 Internet Protocol Addressing

The Internet Network Layer

Host and router network layer functions:



IP datagram format



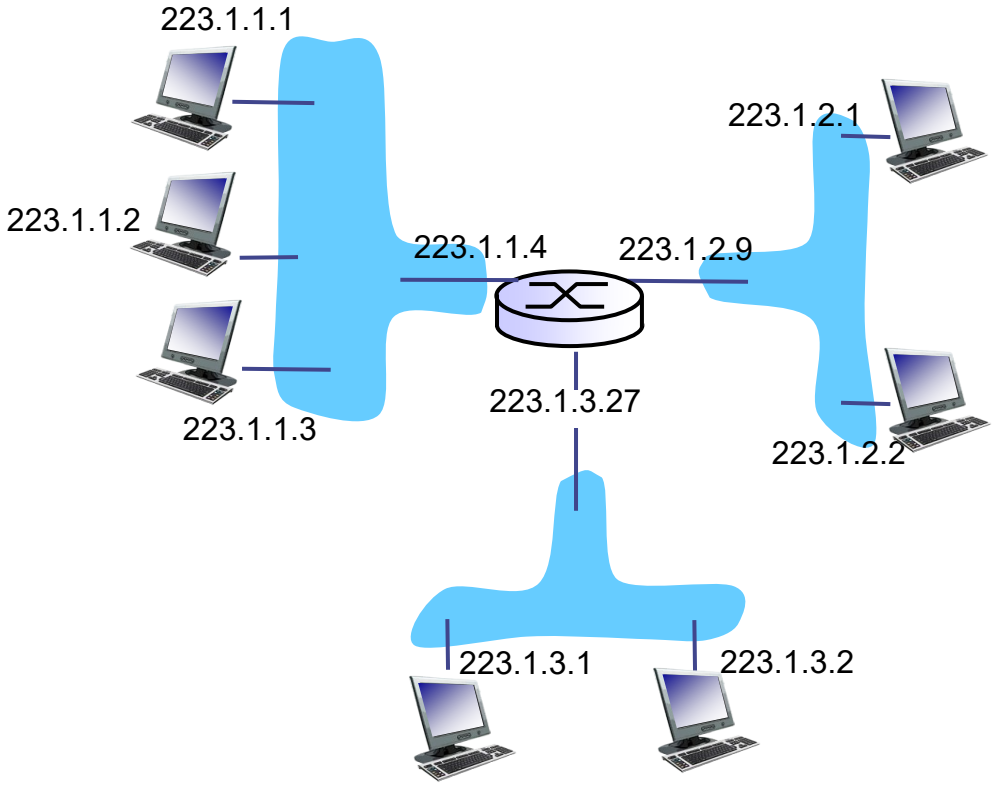
how much overhead?

- 20 bytes of TCP
- 20 bytes of IP
- = 40 bytes + app layer overhead

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IP Addressing - Again

- *IP address:* 32-bit identifier for host or router *interface*
- *interface:* connection between host/router and physical link
  - router's typically have multiple interfaces
  - host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)
- *IP addresses associated with each interface*

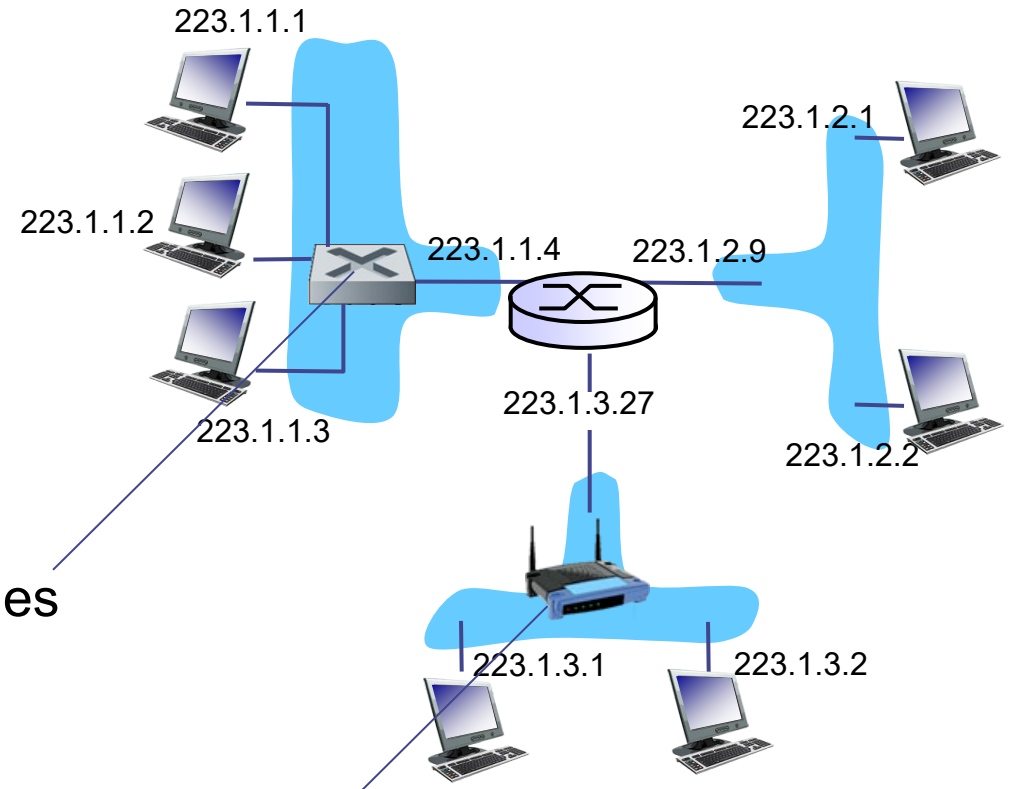


223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1

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IP Addressing: Interfaces



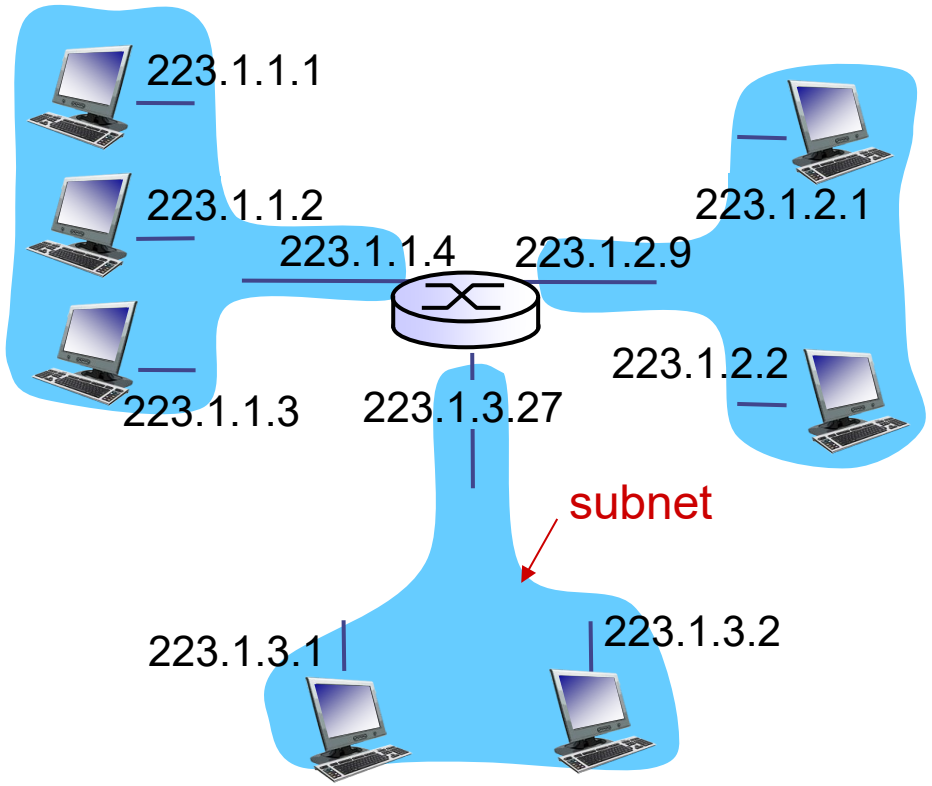
A: wired Ethernet interfaces connected by Ethernet switches

A: wireless WiFi interfaces connected by WiFi base station

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IP Addressing: Subnets

- IP address:
  - subnet part - high order bits
  - host part - low order bits
- *What's a subnet ?*
  - device interfaces with same subnet part of IP address
  - can physically reach each other *without intervening router*

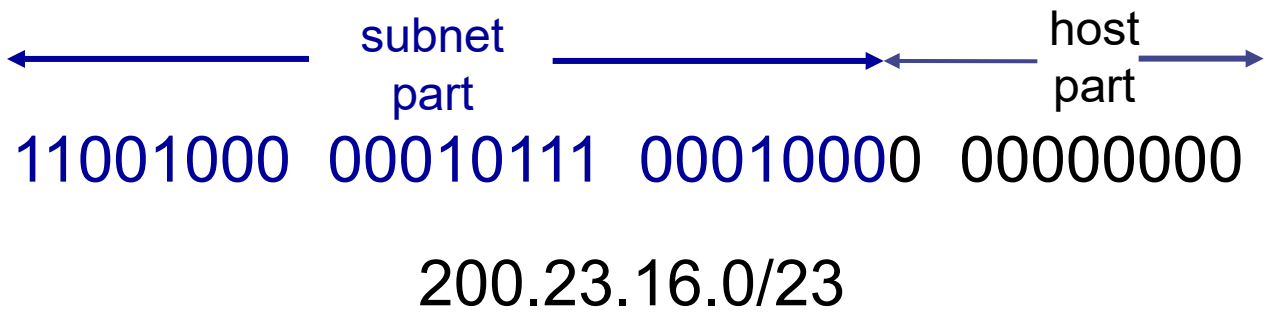


network consisting of 3 subnets

IP Addressing: CIDR

CIDR: Classless InterDomain Routing

- subnet portion of address of arbitrary length
- address format: a.b.c.d/x, where x is # bits in subnet portion of address

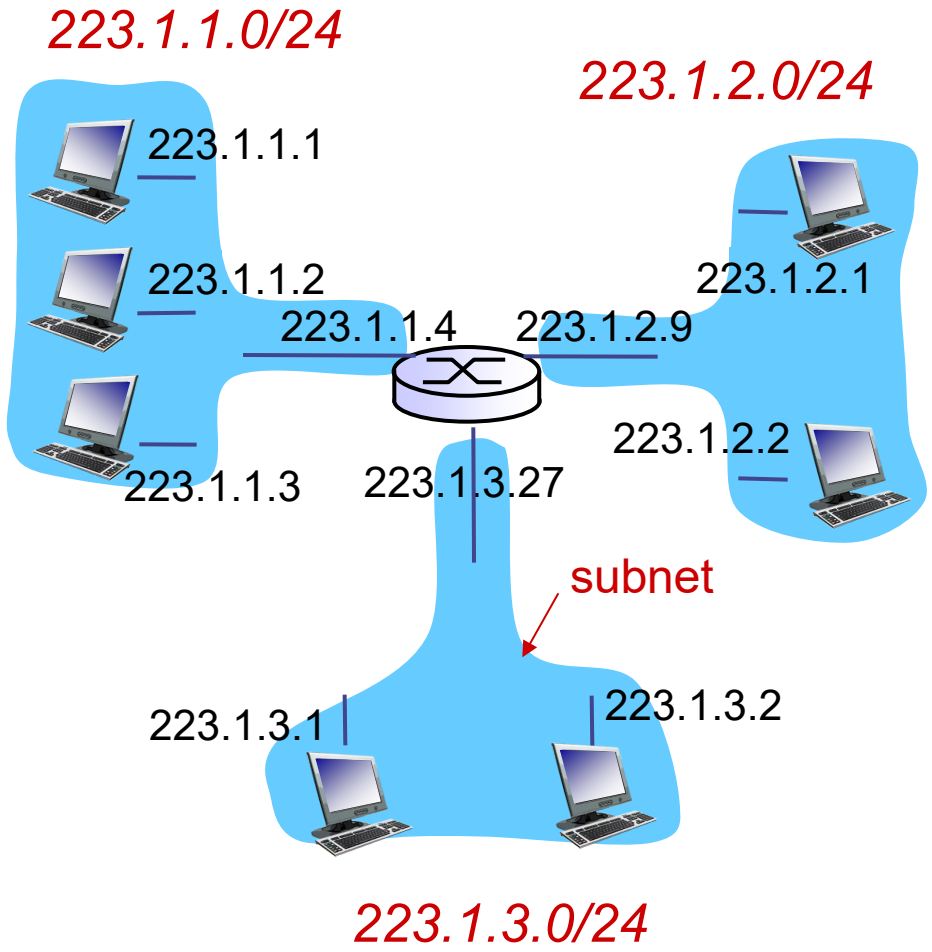


200.23.16.0/23

IP Addressing: Subnets

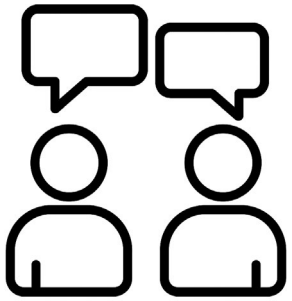
Subnets with CIDR

- To determine the subnets, detach each interface from its host or router, creating islands of isolated networks
- Each isolated network is called a *subnet*

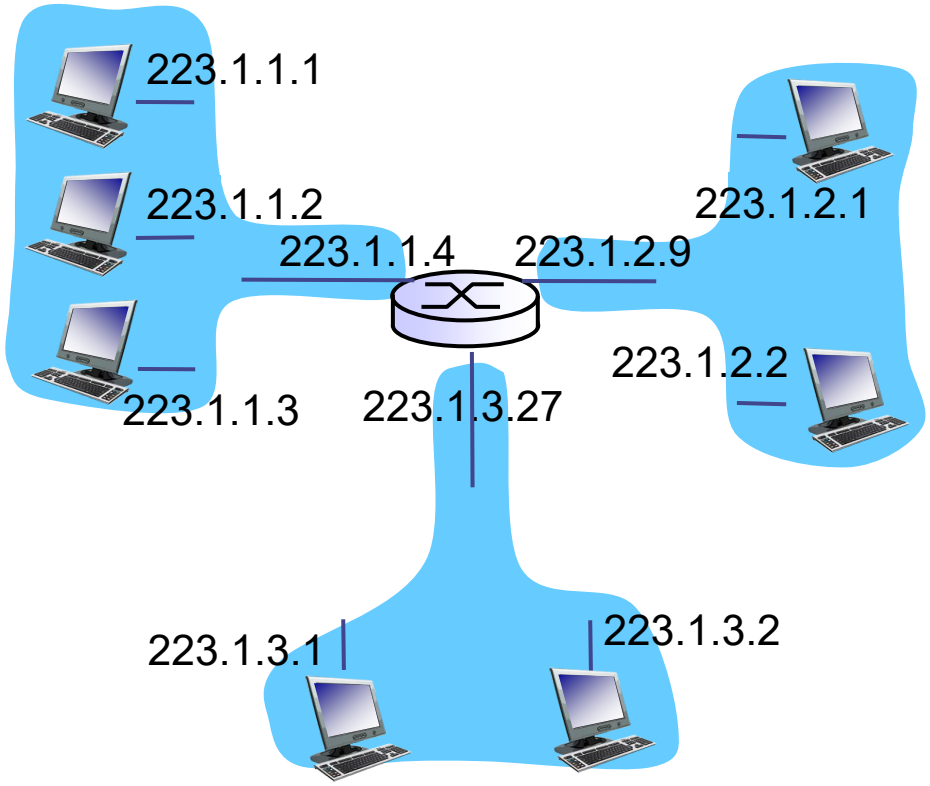


subnet mask: /24

Subnets again...



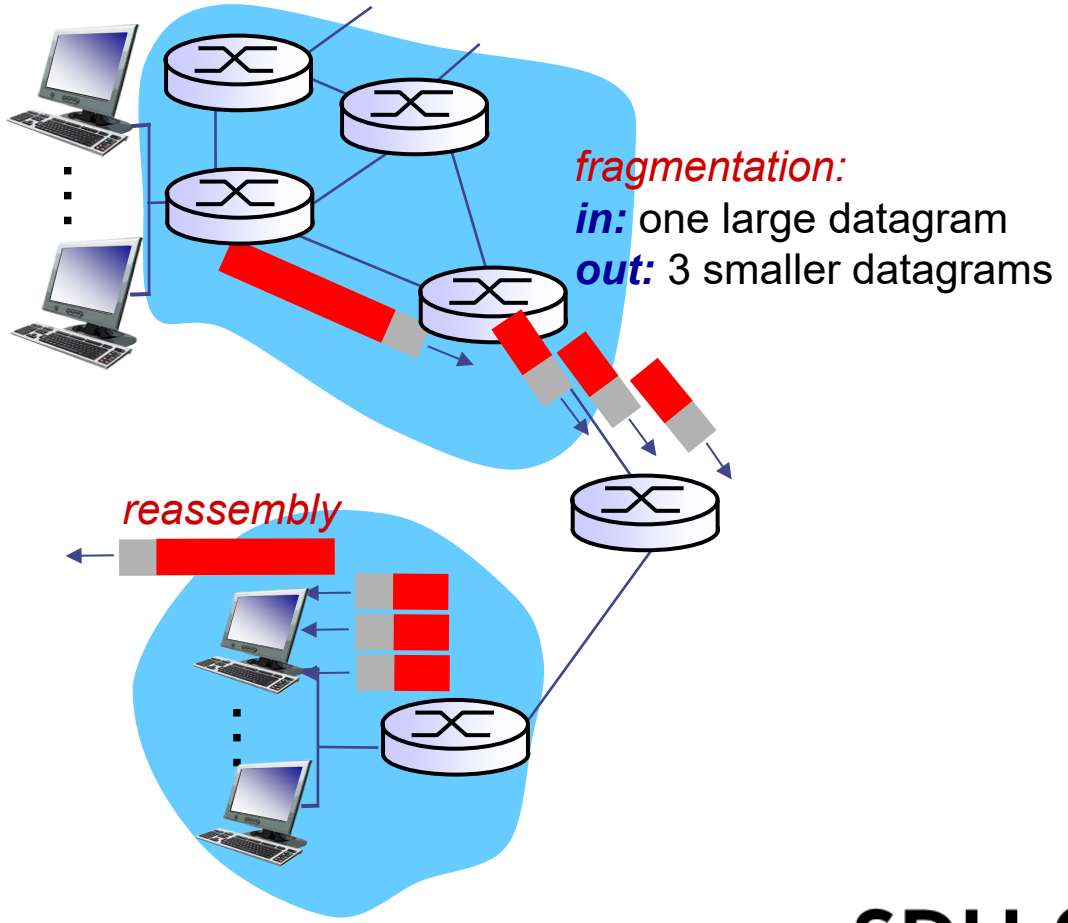
How many subnets if we change the subnet mask to: /16?



IP fragmentation, reassembly

- network links have MTU (max transfer unit size) - largest possible link-level frame
  - different link types, different MTUs

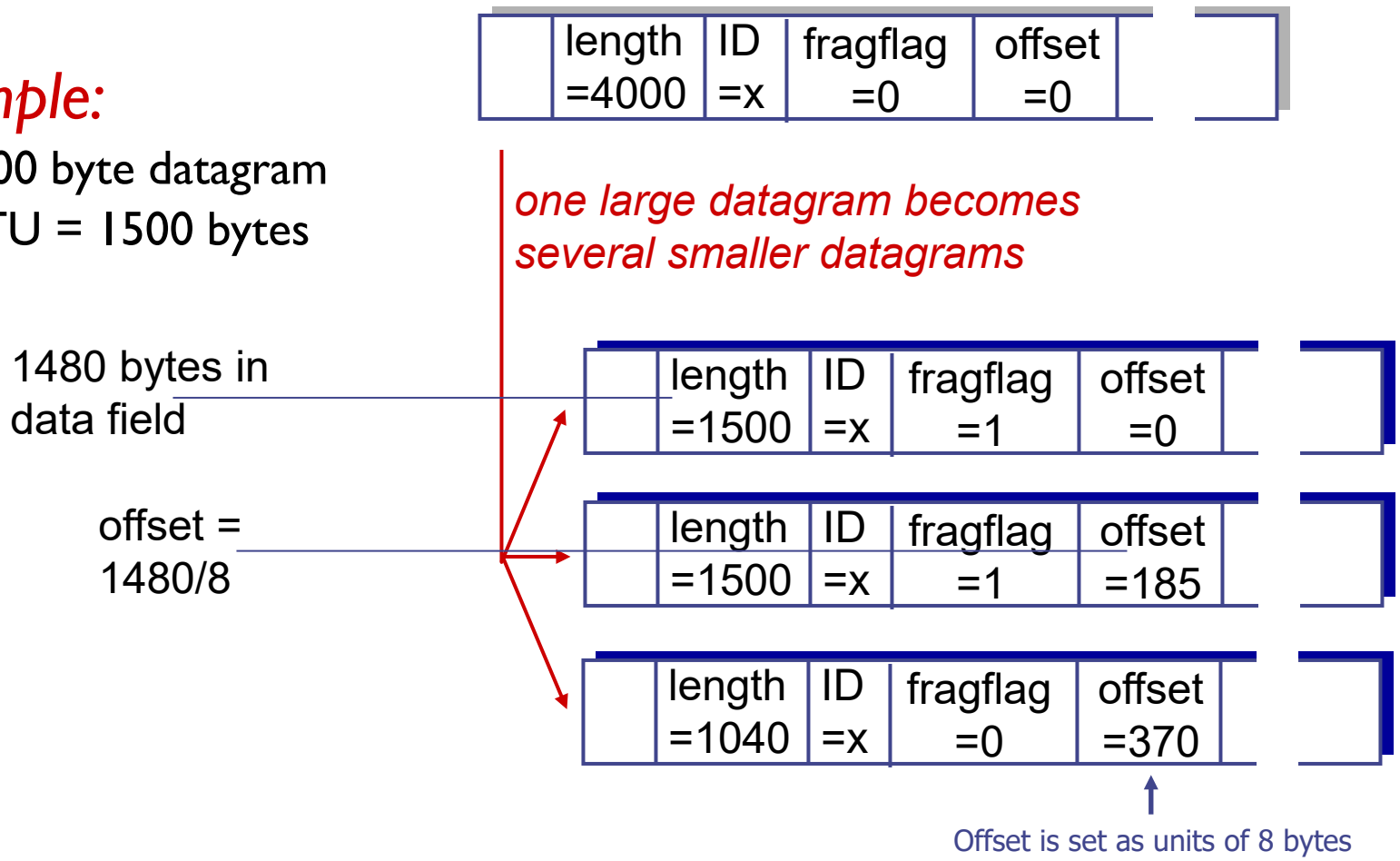
- large IP datagram are divided ("fragmented")
  - one datagram becomes several datagrams
  - "reassembled" only at final destination
  - IP header bits used to identify, order related fragments



IP Fragmentation & Reassembly

example:

- 4000 byte datagram
- MTU = 1500 bytes



IP addresses: How to get one?

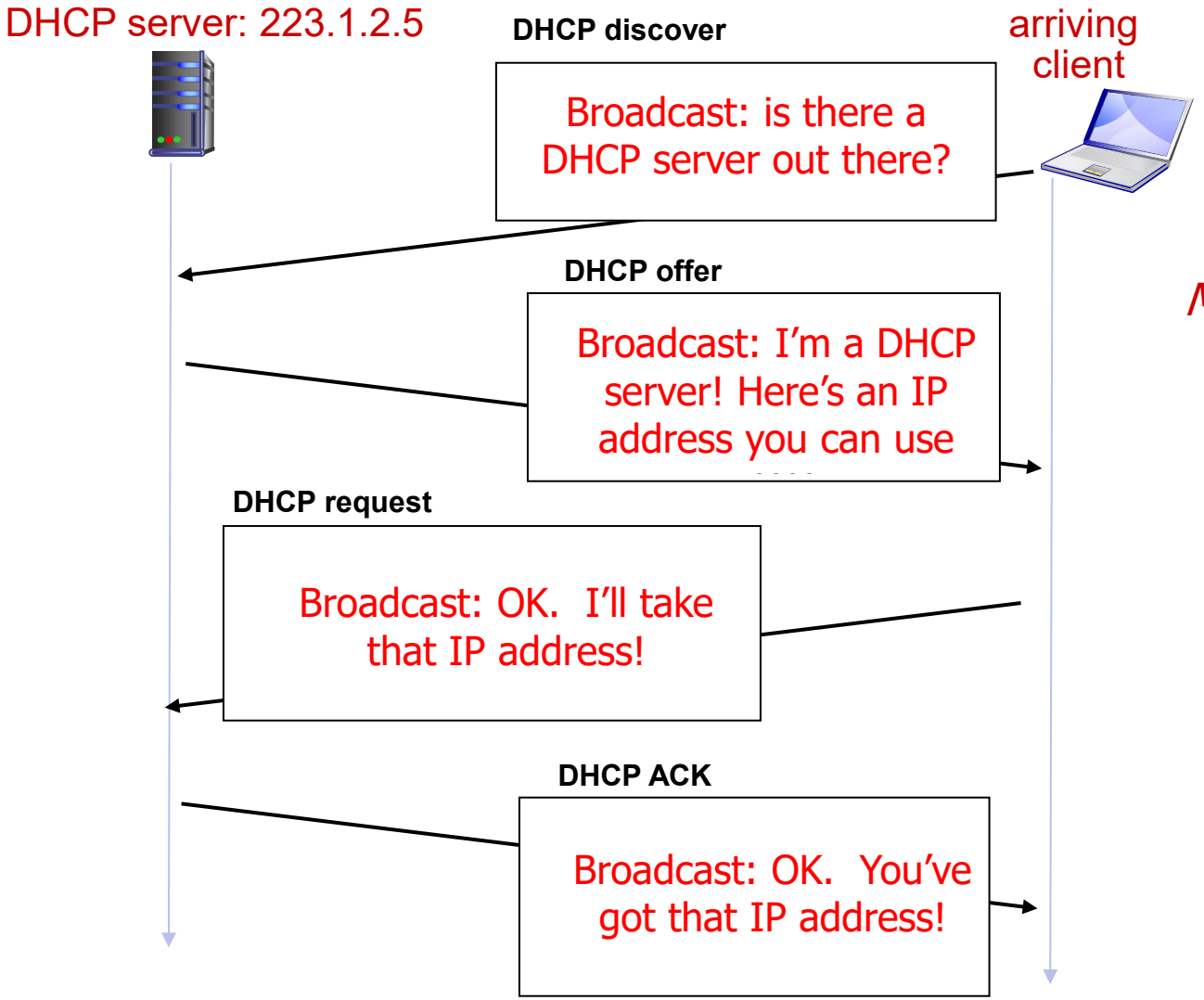
- Hard-coded by system admin in a file
- DHCP: **D**ynamic **H**ost **C**onfiguration **P**rotocol: dynamically get address from as server
  - "plug-and-play"

Goal: allow host to *dynamically* obtain its IP address from network server when it joins network.

- can renew its lease on address in use
- allows reuse of addresses (only hold address while connected/"on")
- support for mobile users who want to join network

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DHCP client-server scenario



Message overview:

- host broadcasts "DHCP discover" msg [optional]
- DHCP server responds with "DHCP offer" msg [optional]
- host requests IP address: "DHCP request" msg
- DHCP server sends address: "DHCP ack" msg

DHCP: more than IP addresses

DHCP can return more than just an allocated IP address on subnet:

- address of first-hop router for client
- name and IP address of DNS sever
- network mask (indicating network versus host portion of address)

IP addresses: how to get one?

Q: How does a *network* get the subnet part of its IP address?

A: Gets the allocated portion of its provider ISP's address space

ISP's block	11001000 00010111 00010000	00000000	200.23.16.0/20
Organization 0	11001000 00010111 00010000	00000000	200.23.16.0/23
Organization 1	11001000 00010111 00010010	00000000	200.23.18.0/23
Organization 2	11001000 00010111 00010100	00000000	200.23.20.0/23
...	.....	....	....
Organization 7	11001000 00010111 00011110	00000000	200.23.30.0/23

IP addressing: the last word...

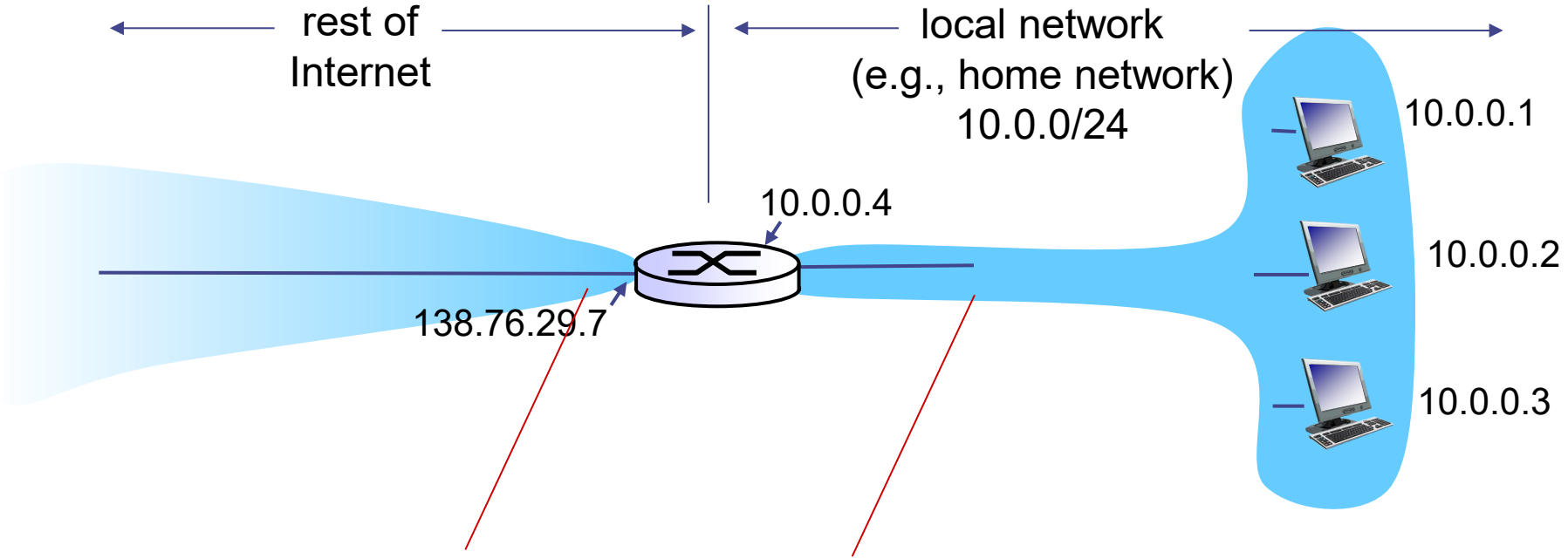
Q: How does an ISP get block of addresses?

A: ICANN: Internet Corporation for Assigned Names and Numbers https://www.icann.org/

- allocates addresses
- manages DNS
- assigns domain names, resolves disputes

7.4 Beyond IPv4: NAT and IPv6

NAT: Network Address Translation



all datagrams *leaving* local network have *same* single source NAT IP address: 138.76.29.7, different source port numbers

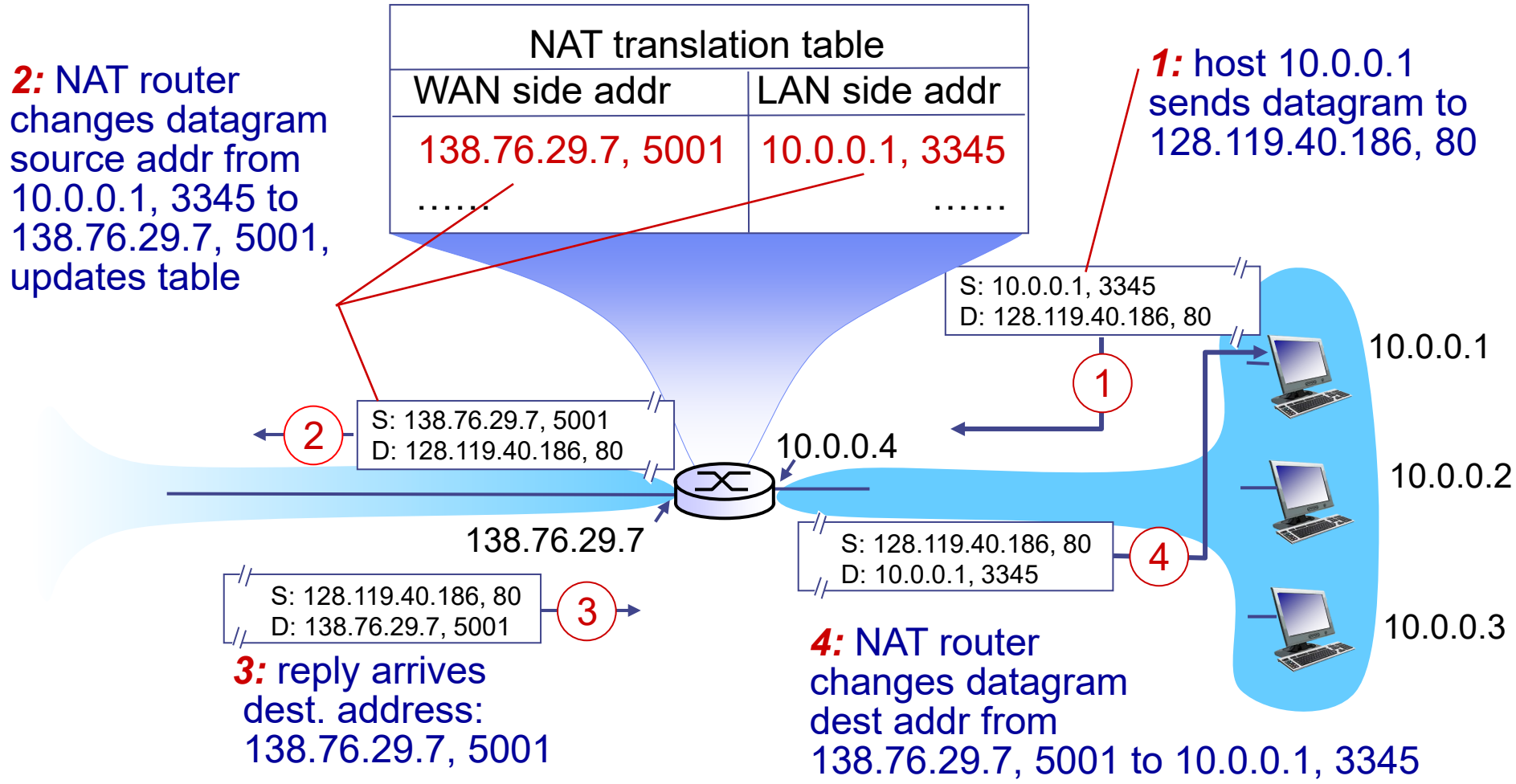
datagrams with source or destination in this network have 10.0.0/24 address for source, destination (as usual)

NAT: Network Address Translation

Motivation: Local network uses just one IP address as far as outside world is concerned:

- Range of addresses not needed from ISP: just one IP address for all devices
- Can change addresses of devices in local network without notifying outside world
- Can change ISP without changing addresses of devices in local network
- Devices inside local net not explicitly addressable, visible by outside world (a security plus)

NAT: Network Address Translation



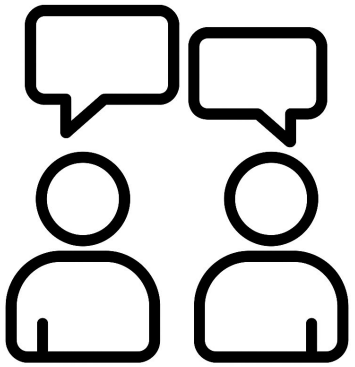
* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/

NAT: Network Address Translation

- *implementation*: NAT router must:
  - *outgoing datagrams: replace* (source IP address, port #) of every outgoing datagram to (NAT IP address, new port #)
    - . . . remote clients/servers will respond using (NAT IP address, new port #) as destination address
  - *remember (in NAT translation table)* every (source IP address, port #) to (NAT IP address, new port #) translation pair
  - *incoming datagrams: replace* (NAT IP address, new port #) in dest fields of every incoming datagram with corresponding (source IP address, port #) stored in NAT table

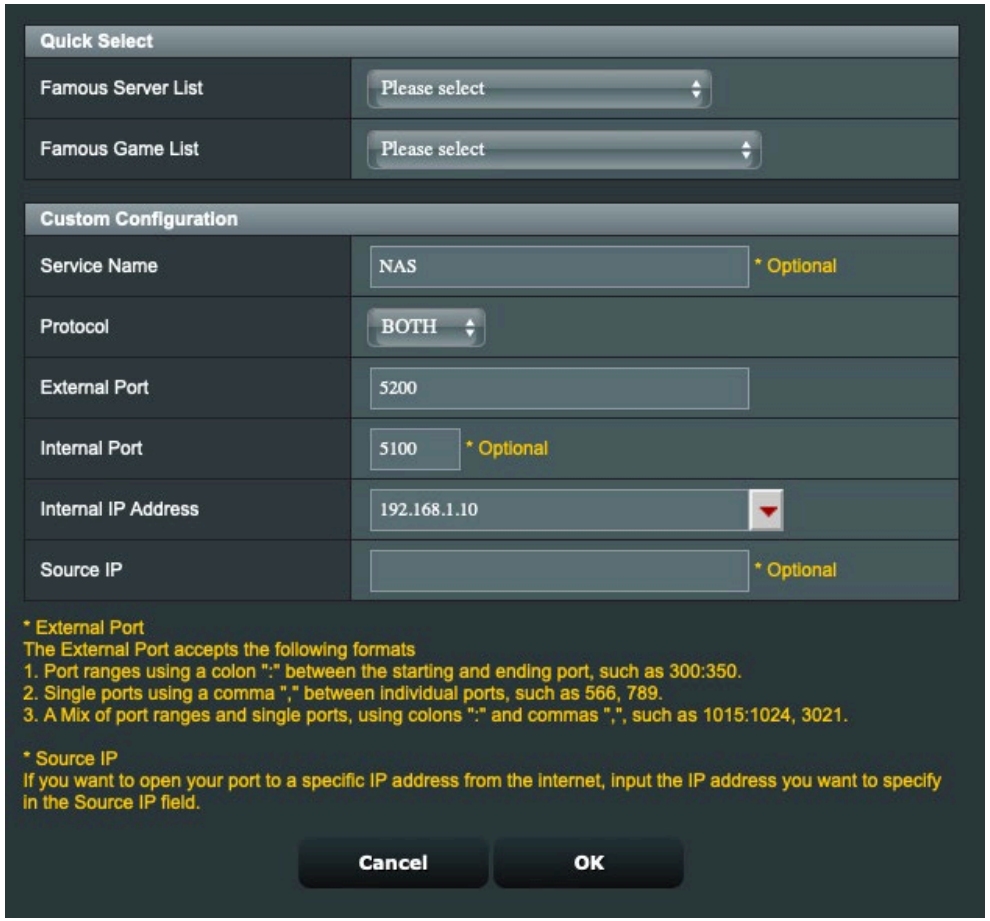
NAT: Network Address Translation

- 16-bit port-number field:
  - 60,000 simultaneous connections with a single LAN-side address!
- NAT is controversial:
  - Routers should only process up to layer 3
  - Address shortage should be solved by IPv6
  - Violates end-to-end argument
    - NAT possibility must be taken into account by app designers, e.g., P2P applications



What if a client wants to connect to server behind NAT?

Port Forwarding– Direct Route



IPv6: Motivation

- 32-bit address space completely allocated[1]
- Additional motivation:
  - header format helps speed processing/forwarding (see format)
  - header changes to facilitate Quality of Service (label traffic types and flows)

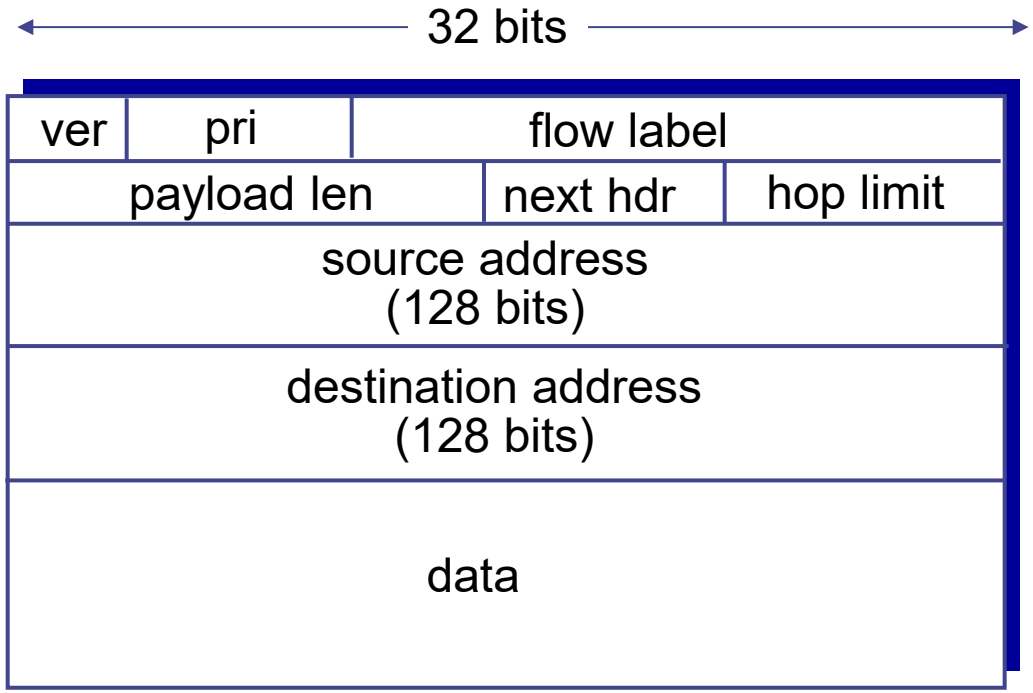
IPv6 datagram format:

  - fixed-length 40 byte header
  - no fragmentation allowed
  - no checksum
  - no options
- *ICMPv6:* new version of ICMP
  - additional message types, e.g. "Packet Too Big"
  - multicast group management functions

[1]https://www.ripe.net/manage-ips-and-asns/ipv4/ipv4-run-out/

IPv6 Datagram Format

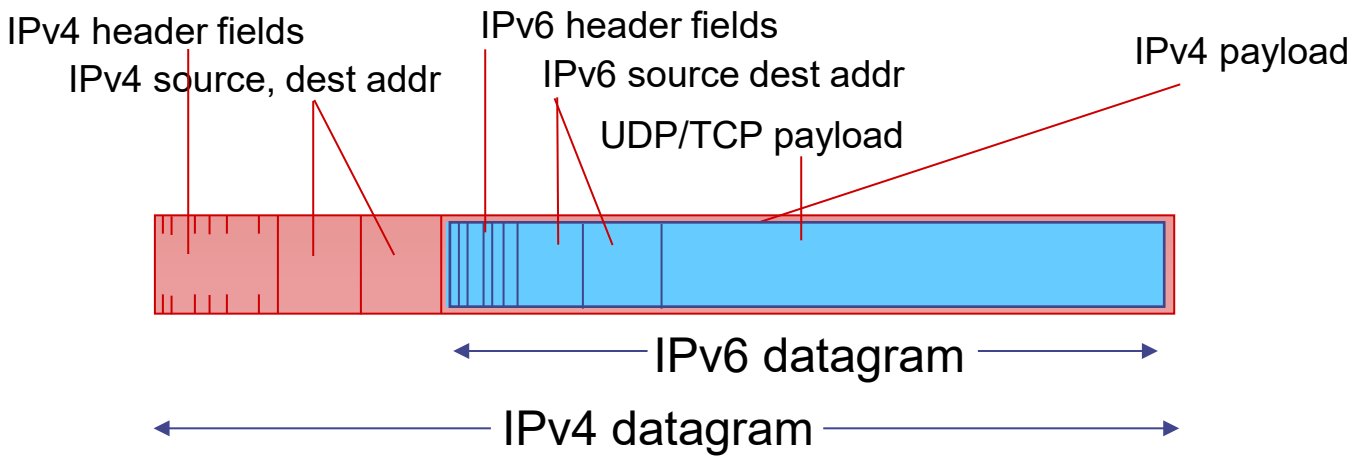
priority: identify priority among datagrams in flow (also known as "Traffic class")
flow Label: identify datagrams in same "flow." (concept of"flow" not well defined).
next header: identify upper layer protocol for data (in the payload)



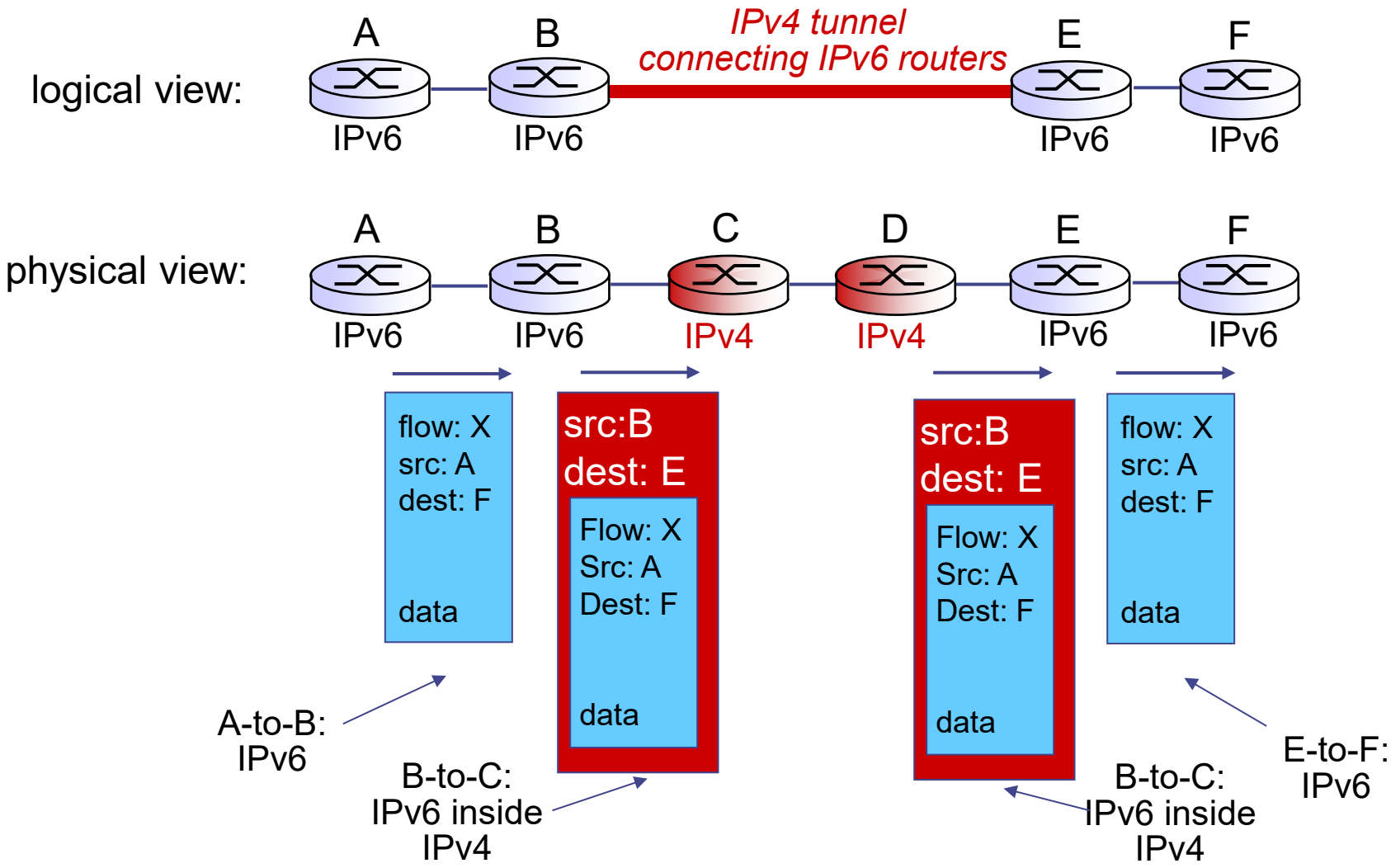
Transitioning from IPv4 to IPv6

- Not all routers can be upgraded simultaneously
  - how will network operate with mixed IPv4 and IPv6 routers?

- *Tunneling:* IPv6 datagram carried as *payload* in IPv4 datagram among IPv4 routers



Tunneling


logical view:
A
B
IPv4 tunnel
connecting IPv6 routers
E
F
IPv6
IPv6
IPv6
IPv6
physical view:
A
B
C
D
E
F
IPv6
IPv6
IPv4
IPv4
IPv6
IPv6
flow: X
src: A
dest: F
data
src:B
dest: E
Flow: X
Src: A
Dest: F
data
src:B
dest: E
Flow: X
Src: A
Dest: F
data
flow: X
src: A
dest: F
data
A-to-B:
IPv6
B-to-C:
IPv6 inside
IPv4
B-to-C:
IPv6 inside
IPv4
E-to-F:
IPv6

IPv6: adoption

- Google: approx. 50% of clients access services via IPv6

- *Long (long!) time for deployment, use*
  - 25 years and counting!
  - think of application-level changes in last 25 years: WWW, social media, streaming media, gaming, telepresence, …
  - *Why?*

Today we introduce the first hand-in assignment of this course, you must hand in a journal of your work with Wireshark Lab. 6 in the hand-in folder at itsLearning.
Deadline is 16/12-2026.
Document with screenshots and answers to the questions.

Be aware of the difference between Unix systems and Windows for this Wireshark Lab exercise.